SURYANSH KUMAR PATHAK

Agentic & Generative AI Engineer

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LinkedIn | GitHub | LeetCode | Portfolio

PROFESSIONAL SUMMARY

Computer Science Engineering student specializing in Generative AI, Large Language Models and Retrieval-Augmented Generation, with a foundation in Machine Learning, Deep Learning and NLP. Develop RAG applications using Python, LangChain, Hugging Face, vector databases and FastAPI, and deploy them with Docker. Experience with LLM fine-tuning using LoRA and QLoRA, and with Ragas-based evaluation to measure retrieval quality. Focused on reliability, latency and production readiness.

EDUCATION

B.Tech, Computer Science & Engineering — CGPA: 7.00 / 10.00 2023 – 2027 (Expected)
Shri Ramswaroop Memorial University | Lucknow, Uttar Pradesh, India

Intermediate (Class XII), Science (PCM) 2022 – 2023
M.V.V.M. School | Ballia, Uttar Pradesh, India

High School (Class X) 2020 – 2021
C.B Inter College | Ballia, Uttar Pradesh, India

TECHNICAL SKILLS

Languages: Python, C++, SQL

AI / ML & GenAI: LangChain, LangGraph, RAG, tool integration, agent workflows, Scikit-learn, TensorFlow, Keras, Hugging Face, vector database (ChromaDB, FAISS), Ollama

Evaluation: Ragas (retrieval precision, faithfulness), evaluation harnesses, latency and error observability dashboards

Databases & Vector DBs: SQLite, ChromaDB, FAISS, BM25 (rank_bm25), semantic search

Backend & Deployment: FastAPI, REST APIs, JSON, Streamlit, Docker (basic), Render deployment, Git, GitHub

PROJECTS

Multimodal RAG Application — *Image, Document & Table Question-Answering*

- Built a multimodal RAG application on Groq models that answers questions over images, documents and tables, with a Claude-style chat interface and an attach control for uploading files
- Architected a decoupled system with a FastAPI backend serving the RAG pipeline and a Streamlit frontend for the chat UI, and deployed the application on Render
- Implemented the RAG pipeline with LangChain Runnables (LCEL), RecursiveCharacterTextSplitter chunking and Gemini embeddings (gemini-embedding-2) for retrieval
- Built an automated evaluation harness with 19 test cases covering tool use, safety, memory, failure handling and file retrieval; reached a 100% pass rate and 100% tool-selection F1, with 12.9 s median (36.0 s p95) answer latency

Tools: *Python, FastAPI, Streamlit, LangChain, Groq, Gemini Embeddings, Render, Git*

Customer Support RAG Assistant

- Designed a customer-support Q&A assistant with hybrid retrieval (BM25 + ChromaDB vector search) and Cross-Encoder re-ranking to sharpen context selection
- Implemented the pipeline with LangChain LCEL and Hugging Face BAAI/bge-m3 embeddings, generating responses with Groq openai/gpt-oss-20b
- Logged every query to a JSONL observability store and surfaced it in a Streamlit dashboard for monitoring intents, latency and errors
- Added Ragas evaluation and measured 6.19 s average total latency and 4.58 s average retrieval + re-ranking latency across 16 queries, with 2 recorded errors

Tools: Python, LangChain (LCEL), ChromaDB, Cross-Encoder Reranker, Hugging Face, Groq, Ragas, Streamlit

AI Medical Assistant

- Developed an LLM-powered assistant that analyzes symptoms and returns preliminary condition predictions with recommendations
- Grounded responses through a RAG pipeline over a vector database, exposed via a FastAPI REST backend
- Containerized the application with Docker and deployed it as a scalable service

Tools: Python, Transformers, LangChain, FastAPI, Docker

KEY ACHIEVEMENTS

- Selected among the top teams in the college-level round of Smart India Hackathon for an AI-driven solution to a real-world problem
- Solved 250+ LeetCode problems across data structures and algorithms
- Earned L&T certification as a Data Science Specialist